

NEW MCQs from: عشر درجات قبل

الميد.pdf (pdf1_arabic.txt) - Set 2

Instructions: Choose the best answer for each question. These questions are different from the previous set.

1. Which logic system is First-Order Logic (FOL) considered an extension of? A) Modal Logic B) Temporal Logic C) Propositional Logic D) Description Logic **Correct Answer: C**
2. What fundamental assumption does Propositional Logic make about the world, according to the document? A) The world contains objects and relations. B) The world contains facts. C) The world is constantly changing. D) The world can be modeled by graphs. **Correct Answer: B**
3. In the FOL representation `friends(Arun, Vijay)`, what does `friends` represent? A) A constant B) A variable C) A function D) A predicate (representing a relationship) **Correct Answer: D**
4. Which of these is listed as an example of a Function in the basic elements of FOL? A) Brother B) Pune C) x D) Sqrt **Correct Answer: D**
5. Which quantifier is associated with the concept of being "True for at least one"? A) $\forall$ (Universal) B) $\exists$ (Existential) C) $\neg$ (Negation) D) $\rightarrow$ (Implication) **Correct Answer: B**
6. What is the typical logical connective paired with the Existential Quantifier ($\exists$)? A) Implication ($\rightarrow$) B) Disjunction ($\vee$) C) Conjunction ($\wedge$) D) Equality ($=$) **Correct Answer: C**
7. What is the typical logical connective paired with the Universal Quantifier ($\forall$)? A) Conjunction ($\wedge$) B) Disjunction ($\vee$) C) Implication ($\rightarrow$) D) Biconditional ($\leftrightarrow$) **Correct Answer: C**
8. Consider the property of quantifiers: Is $\exists x \exists y P(x, y)$ logically equivalent to $\exists y \exists x P(x, y)$? A) Yes, always. B) No, never. C) Only if P is symmetric. D) Only if the domain is infinite. **Correct Answer: A**

9. The statement $\forall x \text{ likes}(x, \text{IceCream})$ is equivalent via quantifier duality to which other statement? A) $\exists x \neg \text{likes}(x, \text{IceCream})$ B) $\neg \exists x \neg \text{likes}(x, \text{IceCream})$ C) $\neg \forall x \text{ likes}(x, \text{IceCream})$ D) $\exists x \text{ likes}(x, \text{IceCream})$ **Correct Answer: B**
10. The statement $\exists x \text{ plays}(x, \text{Cricket})$ is equivalent via quantifier duality to which other statement? A) $\forall x \neg \text{plays}(x, \text{Cricket})$ B) $\neg \forall x \text{ plays}(x, \text{Cricket})$ C) $\neg \forall x \neg \text{plays}(x, \text{Cricket})$ D) $\forall x \text{ plays}(x, \text{Cricket})$ **Correct Answer: C**
11. How is the statement "Virat loves Anushka" represented in FOL? A) $\forall x \text{ loves}(\text{Virat}, x)$ B) $\exists x \text{ loves}(x, \text{Anushka})$ C) $\text{loves}(\text{Virat}, \text{Anushka})$ D) $\text{loves}(\text{Anushka}, \text{Virat})$ **Correct Answer: C**
12. If the domain D contains only humans, how is "Mary loves everyone" represented? A) $\forall x (\text{person}(x) \rightarrow \text{love}(\text{Mary}, x))$ B) $\exists x \text{ love}(\text{Mary}, x)$ C) $\forall x \text{ love}(\text{Mary}, x)$ D) $\text{love}(\text{Mary}, \text{person})$ **Correct Answer: C**
13. If the domain D contains humans and non-humans, why is $\forall x \text{ love}(\text{Mary}, x)$ potentially problematic for "Mary loves everyone [human]"? A) It uses the wrong quantifier. B) It implies Mary loves non-human entities as well. C) It uses the wrong predicate. D) It should use conjunction instead of implication. **Correct Answer: B**
14. Which FOL representation correctly captures "No one talks"? A) $\exists x \text{ talk}(x)$ B) $\forall x \text{ talk}(x)$ C) $\neg \exists x \text{ talk}(x)$ (or $\forall x \neg \text{talk}(x)$) D) $\text{talk}(\neg \text{Someone})$ **Correct Answer: C**
15. How is the concept "Everyone loves himself" represented in FOL? A) $\exists x \text{ love}(x, x)$ B) $\forall x \exists y \text{ love}(x, y)$ C) $\forall x \text{ love}(x, x)$ D) $\text{love}(\text{Everyone}, \text{Everyone})$ **Correct Answer: C**
16. What does the FOL sentence $\forall x \forall y \text{ love}(x, y)$ represent? A) Someone loves someone. B) Everyone loves someone. C) Everyone loves himself. D) Everyone loves everyone. **Correct Answer: D**
17. What is the meaning of the FOL sentence $\forall x ((\text{student}(x) \ \& \ x \neq \text{George}) \rightarrow \text{smile}(x))$? A) Only George the student smiles. B) All students smile, including George. C) Every student who is not George smiles. D) No student except George smiles. **Correct Answer: C**
18. In the expert system workflow described, what component processes the user query using rules and facts? A) User Interface B) Knowledge Base C) Inference Engine D) Explanation System **Correct Answer: C**
19. What happens in the expert system workflow if the Inference Engine needs additional information? A) It makes an assumption. B) It consults the Explanation System. C) It asks the user via the User Interface. D) It terminates the process. **Correct Answer: C**

20. What component provides the reasoning behind the conclusion reached by the expert system? A) Knowledge Base B) Inference Engine C) User Interface D) Explanation System **Correct Answer: D**
21. What is the difference between a problem domain and a knowledge domain for an expert? A) They are the same thing. B) Problem domain is where the expert works; knowledge domain is the expertise used. C) Knowledge domain is broader than the problem domain. D) Problem domain involves facts; knowledge domain involves rules. **Correct Answer: B**
22. What are the two kinds of inference mentioned for simple rule-based systems? A) Deductive and Inductive B) Abductive and Analogical C) Forward chaining and Backward chaining D) Probabilistic and Fuzzy **Correct Answer: C**
23. In Forward Chaining, what typically triggers a rule to perform its action? A) A user query. B) The system reaching a predefined goal. C) Input data in working memory matching the rule's conditions. D) A request from the explanation system. **Correct Answer: C**
24. Forward chaining is also called data-directed inference because: A) It requires large amounts of data. B) Inferences are triggered by the arrival of new data. C) It directs the user to input data. D) It primarily manipulates data structures. **Correct Answer: B**
25. What is the state of the working memory at the beginning of the forward chaining medical example? A) Empty B) Contains only rules C) Contains: runny nose, temperature=104F, headache, cough D) Contains the final diagnosis: influenza **Correct Answer: C**
26. In the forward chaining medical example, which rule allows the system to assert (fever) ? A) R1: if (nasal congestion and virosis)... B) R2: if (runny nose)... C) R4: if (temp > 100)... D) R6: if (fever and itchiness and cough)... **Correct Answer: C** (Because temperature = 104F > 100)
27. In the forward chaining medical example, which rule allows the system to assert (itchiness) based on the initial facts? A) R2: if (runny nose)... B) R3: if (body aches)... C) R5: if (headache)... D) R6: if (fever and itchiness and cough)... **Correct Answer: C** (Because headache is an initial fact)
28. What condition must be met for rule R6 (if (fever and itchiness and cough) , then assert (virosis)) to fire in the forward chaining example? A) Only fever must be in working memory. B) fever , itchiness , and cough

must all be in working memory. C) virosis must already be in working memory.

D) The goal must be virosis . **Correct Answer: B**

29. What is the final action performed by rule R1 in the forward chaining example? A) Assert (nasal congestion) B) Assert (virosis) C) Diagnose (influenza) and exit D) Ask the user for more symptoms **Correct Answer: C**
30. Backward chaining is also known as goal-directed inference because: A) It always reaches the correct goal. B) It starts with data and works towards a goal. C) Inferences are performed to prove a specific goal or hypothesis. D) It directs the user towards a goal. **Correct Answer: C**
31. In backward chaining, if the system needs to know the value of 'X', what does it search for first? A) Facts in working memory that match 'X'. B) Rules whose conditions mention 'X'. C) Rules whose conclusions mention 'X'. D) User input related to 'X'. **Correct Answer: C**
32. In the backward chaining medical example, what triggers the initial firing of R1? A) The presence of nasal congestion in working memory. B) The goal diagnosis(influenza) matches the conclusion of R1. C) The presence of runny nose in working memory. D) The system always starts with R1. **Correct Answer: B**
33. When R1 fires in the backward chaining example, what become the new subgoals? A) runny nose and fever B) nasal congestion and virosis C) headache and cough D) fever, itchiness, and cough **Correct Answer: B**
34. How is the subgoal (nasal congestion) proven in the backward chaining example? A) It is an initial fact. B) Rule R2 fires, creating subgoal (runny nose), which is found in working memory. C) Rule R1 concludes it. D) The user confirms it. **Correct Answer: B**
35. When trying to prove the subgoal (virosis) in backward chaining, rule R6 fires. What are the new subgoals created by R6? A) nasal congestion B) runny nose, headache, cough C) fever, itchiness, cough D) temperature > 100 **Correct Answer: C**
36. How is the subgoal (fever) (needed for R6) proven in the backward chaining example? A) It is an initial fact. B) Rule R4 fires, creating subgoal (temperature > 100), which is true based on working memory (temperature = 104). C) Rule R5 concludes it. D) It fails to be proven. **Correct Answer: B**

37. Why does the attempt to use rule R3 (if (body aches) , then assert (itchiness)) fail when trying to prove (itchiness) in the backward chaining example? A) itchiness was already proven by R5. B) R3 is an invalid rule. C) The subgoal (body aches) cannot be proven from working memory or other rules. D) The system decided to use R5 instead. **Correct Answer: C**
38. How is the subgoal (itchiness) (needed for R6) eventually proven in the backward chaining example? A) It is an initial fact. B) Rule R3 fires successfully. C) Rule R5 fires, creating subgoal (headache) , which is found in working memory. D) It is assumed to be true. **Correct Answer: C**
39. How is the subgoal (cough) (needed for R6) proven in the backward chaining example? A) Rule R1 concludes it. B) Rule R6 concludes it. C) It is found directly in the working memory. D) It is inferred from (runny nose) . **Correct Answer: C**
40. What does the successful return of 'true' from all recursive procedures in the backward chaining example signify? A) All rules in the knowledge base are correct. B) The working memory contains all possible facts. C) The initial goal (diagnosis(influenza)) has been successfully proven. D) The inference engine has stopped. **Correct Answer: C**
41. Which FOL element represents placeholders like 'x' or 'a'? A) Constant B) Variable C) Predicate D) Function **Correct Answer: B**
42. Which FOL element represents specific entities like '1' or 'Pune'? A) Constant B) Variable C) Predicate D) Quantifier **Correct Answer: A**
43. Which FOL element represents properties or relations like 'Father' or 'loves'? A) Constant B) Variable C) Predicate D) Connective **Correct Answer: C**
44. Which FOL element maps terms to terms, like 'LeftLegOf'? A) Constant B) Variable C) Predicate D) Function **Correct Answer: D**
45. Which FOL elements include symbols like '^', 'V', '¬'? A) Quantifiers B) Predicates C) Connectives D) Constants **Correct Answer: C**
46. Which FOL elements include symbols like '∀', '∃'? A) Quantifiers B) Predicates C) Connectives D) Functions **Correct Answer: A**
47. FOL allows talking about all bacteria when sterilizing a jar without naming them explicitly. This is achieved primarily through the use of: A) Constants B) Functions C) Predicates D) Quantifiers **Correct Answer: D**

48. Representing "A person is allowed access if authorized OR known to someone with access" requires which FOL features? A) Only predicates and constants. B) Predicates, variables, quantifiers, and connectives (like OR). C) Only functions and equality. D) Only universal quantifiers. **Correct Answer: B**
49. In the forward chaining medical example, the final diagnosis (influenza) is added to: A) The rule base B) The working memory C) The explanation system D) The user interface directly **Correct Answer: B** (implicitly, before the 'exit' action)
50. The process of an inference engine searching rules until an 'IF' clause is known to be true and then adding the 'THEN' clause to the dataset is characteristic of: A) Backward Chaining B) Forward Chaining C) Knowledge Acquisition D) Domain Analysis **Correct Answer: B**
51. The process where a system needs to know a value, searches for rules concluding that value, and then tries to prove the conditions of those rules is characteristic of: A) Backward Chaining B) Forward Chaining C) Fact Verification D) Rule Writing **Correct Answer: A**
52. What component of an expert system holds the domain-specific rules and facts? A) Inference Engine B) Working Memory C) Knowledge Base D) User Interface **Correct Answer: C**
53. What component holds the temporary, case-specific facts during a reasoning session? A) Inference Engine B) Working Memory / Case-Specific Facts C) Knowledge Base D) Explanation System **Correct Answer: B**
54. The duality $\forall x P(x) \equiv \neg \exists x \neg P(x)$ means that a universal statement can be expressed using: A) Only universal quantifiers B) Negation and an existential quantifier C) Only existential quantifiers D) Conjunction and implication **Correct Answer: B**
55. The duality $\exists x P(x) \equiv \neg \forall x \neg P(x)$ means that an existential statement can be expressed using: A) Only existential quantifiers B) Negation and a universal quantifier C) Only universal quantifiers D) Disjunction and equality **Correct Answer: B**
56. Consider $\exists x \forall y \text{ loves}(x, y)$ vs $\forall y \exists x \text{ loves}(x, y)$. The difference highlights that the order of _____ matters. A) Predicates B) Constants C) Connectives D) Quantifiers (when mixed) **Correct Answer: D**
57. If a rule IF X THEN Y fires in forward chaining, Y becomes a new: A) Goal B) Rule C) Fact in working memory D) Query to the user **Correct Answer: C**

58. If a rule IF X THEN Y is used in backward chaining to prove goal Y, X becomes a new: A) Fact in working memory B) Rule C) Subgoal to be proven D) Final answer
Correct Answer: C
59. The FOL representation $\forall x(\text{Student}(x) \wedge x \neq \text{George}) \rightarrow \text{Smiles}(x)$ uses which connectives? A) AND ($\wedge$), NOT EQUAL ($\neq$), IMPLIES ($\rightarrow$) B) OR ($\vee$), EQUAL ($=$), AND ($\wedge$) C) ONLY IMPLIES ($\rightarrow$) D) AND ($\wedge$), OR ($\vee$) **Correct Answer: A**
60. The medical diagnosis example primarily illustrates which type of AI system? A) Natural Language Processing System B) Robotics System C) Rule-Based Expert System D) Machine Learning System **Correct Answer: C**

NEW MCQs from: Integration of Semantic Networks, Frames, and ontologies in ES.pdf (pdf2_integration.txt) - Set 2

Instructions: Choose the best answer for each question. These questions are different from the previous set.

1. What is the central component of an expert system responsible for storing domain-specific information and rules? A) Inference Engine B) User Interface C) Knowledge Base D) Working Memory **Correct Answer: C**
2. The efficiency of problem-solving in an expert system is directly impacted by the choice of: A) Programming language B) Hardware platform C) Knowledge Representation (KR) method D) User interface design **Correct Answer: C**
3. Which inference mechanism is particularly suited for situations where facts are observed and conclusions need to be inferred automatically? A) Backward Chaining B) Forward Chaining C) Abductive Reasoning D) Analogical Reasoning **Correct Answer: B**
4. What is the 'Match' phase in the inference engine cycle? A) Selecting the best rule to fire. B) Executing the action part of a rule. C) Comparing rule conditions (IF parts) with current facts. D) Updating the working memory. **Correct Answer: C**
5. If multiple rules match the current facts, what strategy does the inference engine use? A) Fires all matching rules simultaneously. B) Stops the inference process. C) Applies a conflict resolution strategy (e.g., priority, specificity). D) Asks the user which rule to fire. **Correct Answer: C**

6. The example "Car has-a Engine → Car has-a Part (from Engine)" illustrates which type of rule in semantic networks? A) Inheritance B) Compositional C) Equivalence D) Constraint **Correct Answer: B**
7. What is the purpose of an Equivalence rule in semantic networks? A) To enforce logical correctness. B) To handle synonyms or aliases between concepts. C) To categorize instances. D) To propagate properties via 'is-a' links. **Correct Answer: B**
8. The example "Automobile $\equiv$ Car → Automobile has-a SteeringWheel" demonstrates which rule type? A) Inheritance B) Compositional C) Equivalence D) Classification **Correct Answer: C**
9. Which rule type is used to **enforce logical correctness**, such as "Vehicle with Wings → Must be Aircraft"? A) Classification B) Default/Defeasible C) Associative D) Constraint **Correct Answer: D**
10. What kind of **logic often involves exceptions**, as seen in "Bird → can Fly; but Penguin → can't Fly"? A) Inheritance logic B) Compositional logic **C) Default/Defeasible logic** D) Constraint logic **Correct Answer: C**
11. Associative rules in semantic networks represent what kind of relationships? A) Strict hierarchical links. B) Part-whole structures. **C) Indirect relationships** or co-occurrence. D) Logical necessities. **Correct Answer: C**
12. In forward chaining within semantic networks, the process typically stops when: A) The first rule is fired. B) A specific goal is disproven. **C) No new information can be inferred (saturation) or a goal is achieved.** D) The user interrupts the process. **Correct Answer: C**
13. In the forward chaining example (Cat is-a Mammal...), the step "Apply transitive reasoning: Cat is-an Animal" uses which relationship? A) has-a **B) is-a** (from Cat to Mammal and Mammal to Animal) C) part-of D) associated-with **Correct Answer: B**
14. In the second forward chaining example (Dog), Rule 1 ("IF X is-a Animal THEN X has-a Heart") is applied after which fact is established? A) Dog has-a Tail B) Dog is-a Mammal **C) Dog is-a Animal** (derived via inheritance) D) Animal has-a Heart **Correct Answer: C**
15. In the second forward chaining example (Dog), what is the final conclusion reached by applying Rule 2? A) Dog has-a Heart B) Dog is-a Animal **C) Dog is Candidate for Surgery** D) Dog has-a Tail **Correct Answer: C**
16. In backward chaining, if the goal is "Is Dog a Candidate for Surgery?", what does the system do first? A) Look for facts about Dogs. **B) Look for rules whose conclusion**

matches the goal. C) Start forward chaining from known facts. D) Ask the user if the Dog is a candidate. **Correct Answer: B**

17. In the backward chaining example (Dog), how is the sub-subgoal "Dog is-a Animal" proven? A) It's a direct fact in the network. **B) By tracing the 'is-a' hierarchy: Dog → Mammal → Animal.** C) By applying Rule 1. D) By applying Rule 2. **Correct Answer: B**
18. What is the purpose of Step 1 (Define the Domain and Scope) in the Unified Design Process? A) To capture all possible entities. **B) To clearly specify the boundaries and capabilities of the expert system.** C) To define the relationships between concepts. D) To build the class hierarchy. **Correct Answer: B**
19. Using UML use cases to model expert system roles and interactions is suggested in which step of the Unified Design Process? **A) Step 1: Define Domain and Scope** B) Step 2: Identify Key Concepts C) Step 3: Identify Relationships D) Step 4: Build Hierarchy **Correct Answer: A**
20. What is the primary action in Step 2 (Identify Key Concepts) of the Unified Design Process? A) Writing competency questions. **B) Conducting interviews or analyzing literature to capture domain entities.** C) Defining 'is-a' and 'has-a' links. D) Adding constraints and rules. **Correct Answer: B**
21. Representing captured concepts in a tabular format or CSV is suggested during which step? A) Step 1: Define Domain and Scope **B) Step 2: Identify Key Concepts** C) Step 3: Identify Relationships D) Step 4: Build Hierarchy **Correct Answer: B**
22. Defining formal semantic relationships like 'is-a', 'has-symptom', 'treated-by' occurs in which step? A) Step 2: Identify Key Concepts **B) Step 3: Identify Relationships** C) Step 4: Build Hierarchy D) Step 5: Define Attributes **Correct Answer: B**
23. How might relationships be represented using Frames in Step 3? A) Using directed graph edges. B) Using logical axioms. **C) Using slots, potentially with facets and procedural attachments.** D) Using competency questions. **Correct Answer: C**
24. Building a class tree or using 'is-a' links to create a taxonomy is the main activity of which step? A) Step 3: Identify Relationships **B) Step 4: Build the Class/Frame Hierarchy** C) Step 5: Define Attributes D) Step 6: Define Constraints **Correct Answer: B**
25. Adding metadata like 'hasSeverity (string)' or 'hasAge (int)' to classes/frames occurs in which step? A) Step 4: Build Hierarchy **B) Step 5: Define Attributes / Slots / Properties** C) Step 6: Define Constraints and Rules D) Step 7: Instantiate Individuals **Correct Answer: B**

26. Using OWL/SWRL to add cardinality constraints (e.g., 'hasSymptom min 2') or rules happens in which step? A) Step 4: Build Hierarchy B) Step 5: Define Attributes C) Step 6: Define Constraints and Rules D) Step 7: Instantiate Individuals **Correct Answer: C**
27. Using procedural slot attachments in Frames to enforce constraints is part of which step? A) Step 4: Build Hierarchy B) Step 5: Define Attributes C) Step 6: Define Constraints and Rules D) Step 7: Instantiate Individuals **Correct Answer: C**
28. Providing data-level examples like 'Patient A' or 'PlanA' corresponds to which step? A) Step 5: Define Attributes B) Step 6: Define Constraints C) Step 7: Instantiate Individuals D) Step 8: Validate **Correct Answer: C**
29. Simulating reasoning scenarios and checking if the knowledge base can answer the initial competency questions occurs in which step? A) Step 6: Define Constraints B) Step 7: Instantiate Individuals C) Step 8: Validate Against Competency Questions D) Step 9: Iterate and Refine **Correct Answer: C**
30. Using ontology reasoners like HermiT to find logical conflicts is an action performed during which step? A) Step 2: Identify Concepts B) Step 5: Define Attributes C) Step 8: Validate D) Step 9: Iterate and Refine **Correct Answer: D**
31. Refactoring frame structures or adding missing axioms are activities associated with which step? A) Step 1: Define Scope B) Step 4: Build Hierarchy C) Step 7: Instantiate Individuals D) Step 9: Iterate and Refine **Correct Answer: D**
32. In the Fitness Ontology example, 'Beginner', 'Intermediate', 'Advanced' are instances of which class? A) Goal B) Exercise C) FitnessLevel D) WorkoutPlan **Correct Answer: C**
33. 'Push-ups', 'Squats', 'Running' are instances or types belonging to which class in the Fitness Ontology? A) Goal B) Exercise C) FitnessLevel D) IntensityLevel **Correct Answer: B**
34. Which relationship connects a 'Person' to their 'FitnessLevel' in the Fitness Ontology? A) hasGoal B) assignedPlan C) includesExercise D) hasFitnessLevel **Correct Answer: D**
35. Which relationship connects a 'WorkoutPlan' to its 'IntensityLevel'? A) hasGoal B) assignedPlan C) includesExercise D) hasIntensityLevel **Correct Answer: D**
36. What enables personalized fitness recommendations in the Fitness Ontology example? A) The number of individuals defined. B) The use of forward chaining

only. C) The defined properties (relationships) connecting Person, Goal, Plan, Level, etc. D) The graphical representation. **Correct Answer: C**

37. According to Rule 2 in the Fitness Ontology, what is another condition (besides goal) for assigning Workout Plan A? A) Person is older than 45. B) Person has goal 'Weight Loss'. C) Person is at an advanced fitness level. D) Workout plan includes 'Running'. **Correct Answer: C**
38. Rule 4 in the Fitness Ontology ("If a person is assigned Workout Plan A, then infer goal is build muscle") allows reasoning in which direction? A) From goal to plan assignment. B) From plan assignment back to likely goal. C) From fitness level to plan assignment. D) From exercise type to intensity. **Correct Answer: B**
39. Rule 5 ("If person older than 45, set fitness level to beginner") is an example of inferring what? A) A goal based on age. B) A plan based on age. C) A fitness level based on age. D) An exercise based on age. **Correct Answer: C**
40. In the Fitness Ontology forward chaining example, the goal is to automatically assign PlanA based on what initial fact about JohnDoe? A) JohnDoe is a Person. B) JohnDoe is AdvancedLevel. C) hasGoal(JohnDoe, BuildMuscle). D) includesExercise(PlanA, PushUp). **Correct Answer: C**
41. In the Fitness Ontology backward chaining example, the system looks for a rule whose conclusion matches what target fact? A) assignedPlan(JohnDoe, PlanA) B) hasGoal(JohnDoe, BuildMuscle) C) hasIntensityLevel(PlanA, HighIntensity) D) includesExercise(PlanA, PushUp) **Correct Answer: C**
42. What fact needs to be confirmed to satisfy the condition of Rule 3 in the backward chaining example? A) hasGoal(JohnDoe, BuildMuscle) B) assignedPlan(JohnDoe, PlanA) C) includesExercise(PlanA, PushUp) D) hasFitnessLevel(JohnDoe, AdvancedLevel) **Correct Answer: C**
43. The graphical diagram at the end shows 'john doe' linked to 'Build muscle' via which relationship? A) has Fitness level B) assigned Workout plan C) has goal D) includes exercise **Correct Answer: C**
44. The graphical diagram shows 'Plan A' linked to 'pushup' via which relationship? A) has Intensity level B) assigned Workout plan C) has goal D) includes exercise **Correct Answer: D**
45. The graphical diagram shows 'Plan A' linked to 'high' via which relationship? A) has Intensity level B) assigned Workout plan C) has goal D) includes exercise **Correct Answer: A**

46. Conflict resolution strategies like 'specificity' usually prioritize rules that: A) Were added most recently. B) Have the highest assigned priority number. C) Have more conditions or match a more specific case. D) Are shorter and easier to process. **Correct Answer: C**
47. Conflict resolution strategies like 'recency' usually prioritize rules that: A) Match the most specific facts. B) Match based on facts that were added most recently to working memory. C) Have the fewest conditions. D) Are defined earliest in the rule base. **Correct Answer: B**
48. The process of deriving 'Dog has-a Heart' from 'Dog is-a Mammal' and 'Mammal has-a Heart' is an example of applying which type of rule? A) Compositional B) Constraint C) Inheritance D) Associative **Correct Answer: C**
49. The Unified Design Process aims to bridge the gap between: A) User requirements and system testing. B) Domain understanding and formal knowledge representation. C) Forward chaining and backward chaining. D) Semantic networks and frames. **Correct Answer: B**
50. Competency questions help define which aspect of the expert system during the design process? A) The specific algorithms used by the inference engine. B) The required hardware specifications. C) The scope and intended capabilities of the system. D) The programming language for implementation. **Correct Answer: C**
51. Representing 'Asthma is-a RespiratoryDisease' is an example of which relationship type, crucial for Step 4 (Build Hierarchy)? A) has-symptom B) treated-by C) is-a (subclassing) D) part-of **Correct Answer: C**
52. Adding a rule like 'Disease(?d) ^ hasSymptom(?d, Cough) ^ hasSymptom(?d, Fever) → SuspectPneumonia(?d)' is part of which design step? A) Step 3: Identify Relationships B) Step 4: Build Hierarchy C) Step 5: Define Attributes D) Step 6: Define Constraints and Rules **Correct Answer: D**
53. Testing rule firing using engines like CLIPS or Drools occurs during which design step? A) Step 2: Identify Concepts B) Step 5: Define Attributes C) Step 8: Validate Against Competency Questions D) Step 9: Iterate and Refine **Correct Answer: C**
54. What is the primary difference between forward and backward chaining in terms of initiation? A) Forward starts with rules, backward starts with facts. B) Forward starts with facts/data, backward starts with a goal/hypothesis. C) Forward uses inheritance, backward uses composition. D) Forward is data-driven, backward is rule-driven. **Correct Answer: B**

55. In the context of semantic networks, 'transitive reasoning' is most closely associated with which relationship type? A) has-a B) part-of C) is-a D) causes
Correct Answer: C
56. The 'Act/Fire' phase of the inference engine cycle involves: A) Comparing rules to facts. B) Choosing one rule from matching rules. C) Executing the THEN part of the selected rule. D) Adding initial facts to working memory. **Correct Answer: C**
57. Using an external rule engine like Prolog or CLIPS might be necessary for complex inference with which KR method mentioned in Step 6? A) Ontologies (using OWL/SWRL) B) Frames (with procedural attachments) C) Semantic Networks D) Basic Rule Systems **Correct Answer: C**
58. The example 'Dyspnea' (Shortness of breath) being identified as a 'Symptom' occurs in which design step? A) Step 1: Define Scope B) Step 2: Identify Key Concepts C) Step 3: Identify Relationships D) Step 4: Build Hierarchy **Correct Answer: B**
59. Defining 'hasAge' as an integer datatype property for a class is part of which design step? A) Step 4: Build Hierarchy B) Step 5: Define Attributes / Slots / Properties C) Step 6: Define Constraints D) Step 7: Instantiate Individuals **Correct Answer: B**
60. The overall goal of the Unified Design Process is to create a knowledge representation that is: A) Simple and easy to build quickly. B) Visually appealing. C) Formally structured, consistent, and suitable for reasoning. D) Compatible with only one type of inference engine. **Correct Answer: C**

NEW MCQs from: Semantic Networks _ Lecture 9.pdf (pdf3_semantic_networks.txt) - Set 2

Instructions: Choose the best answer for each question. These questions are different from the previous set.

1. What is the fundamental structure of a semantic network? A) A list of rules. B) A table of facts. **C) A graph with nodes and edges.** D) A hierarchical tree. **Correct Answer: C**
2. Besides organizing knowledge, what other functions do semantic networks support? A) Data storage and backup. B) User interface design. **C) Reasoning,**

inference, and modeling natural language. D) Algorithm optimization. **Correct Answer: C**

3. How do semantic networks facilitate machine processing of information? A) By compressing data. **B) By providing structured organization for interpretation and decision-making.** C) By encrypting information. D) By executing code directly.

Correct Answer: B

4. The relationship "part of" is used to represent what kind of connection? A) Hierarchical classification (is-a) B) Instance relationship (instance-of) C) Possession (has-a) **D) Composition or aggregation (part-whole)** **Correct Answer: D**

5. Nodes in a semantic network can represent: A) Only physical objects. B) Only abstract ideas. **C) Both physical objects and abstract ideas.** D) Only relationships.

Correct Answer: C

6. Are edges in a semantic network typically directional or non-directional? A) Always non-directional. **B) Typically directional, indicating how concepts are connected.** C) Can be either, with no preference. D) Directionality depends on the node type.

Correct Answer: B

7. Which label type denotes that a specific instance belongs to a category (e.g., "Fido" is a "Dog")? A) Is-A B) Part-Of C) Has-A **D) Instance-Of** **Correct Answer: D**

8. Under what condition might an attribute (like "Red") be modeled as a separate node in a semantic network? A) If the attribute is a number. B) If the attribute applies only to one entity. **C) If the attribute is shared across entities or has its own properties.** D) Attributes are never modeled as nodes. **Correct Answer: C**

9. What mechanism allows properties associated with a general concept (parent node) to apply to more specific concepts (child nodes)? A) Semantic Distance B) Labeling **C) Inheritance** D) Node creation **Correct Answer: C**

10. How is semantic distance often measured or estimated in semantic networks? A) By the number of attributes a node has. B) By the complexity of the node label. **C) By the number of edges (path length) between two nodes.** D) By the physical distance on the diagram. **Correct Answer: C**

11. Definitional networks are primarily used for representing what kind of structure? A) Procedural steps B) Specific facts about individuals **C) Hierarchical classifications (taxonomies)** D) Dynamic learning processes **Correct Answer: C**

12. Inferring that dogs are warm-blooded because "Dog is-a Mammal" and mammals are warm-blooded is an application of which type of network? A) Assertional

Network **B) Definitional Network** (due to the hierarchy) C) Executable Network D) Learning Network **Correct Answer: B**

13. What kind of nodes are central to Assertional Networks? A) Nodes representing general concepts/classes. **B) Nodes representing specific individuals or instances.** C) Nodes representing actions or procedures. D) Nodes representing logical implications. **Correct Answer: B**
14. Representing "Patient A has Symptom X" would be typical in which type of semantic network? A) Definitional Network **B) Assertional Network** C) Implicational Network D) Executable Network **Correct Answer: B**
15. What is the main purpose of Implicational Networks? A) To define hierarchies. B) To assert facts about individuals. **C) To represent logical implications for inferring new knowledge.** D) To represent sequences of actions. **Correct Answer: C**
16. Rule-based systems often utilize the principles found in which type of semantic network? A) Assertional Networks B) Definitional Networks **C) Implicational Networks** D) Learning Networks **Correct Answer: C**
17. What type of knowledge is represented by Executable Networks? A) Hierarchical knowledge B) Factual knowledge about instances **C) Procedural knowledge (actions, sequences)** D) Declarative knowledge **Correct Answer: C**
18. Representing the sequence of movements for a robot to assemble a product is an application of which network type? A) Implicational Network B) Definitional Network C) Assertional Network **D) Executable Network** **Correct Answer: D**
19. What distinguishes Learning Networks from other types? A) They only represent hierarchies. B) They are static once created. **C) They are dynamic and evolve based on new data or experiences.** D) They cannot represent relationships. **Correct Answer: C**
20. Recommendation systems that adapt based on user behavior often employ principles of which network type? A) Executable Networks B) Definitional Networks **C) Learning Networks** D) Implicational Networks **Correct Answer: C**
21. What defines a Hybrid Network? A) A network that uses multiple types of nodes. **B) A network that combines elements from two or more other network types (e.g., definitional and assertional).** C) A network that connects to external databases. D) A network designed for both humans and machines. **Correct Answer: B**
22. Representing both the general concept "Dog" and specific instances like "Rex" within the same structure is characteristic of which network type? A) Purely

Definitional B) Purely Assertional **C) Hybrid Network** D) Executable Network

Correct Answer: C

23. In the "Healthy Lifestyle" example, the relationship "If a person exercises, then they feel energetic" is an example of: A) An assertion about an individual. B) A hierarchical definition. **C) An implication.** D) A procedural step. **Correct Answer: C**
24. Following the chain in the "Healthy Lifestyle" example, what is the prerequisite for "having time for exercise"? A) Sleeping early **B) Waking up early** C) Feeling energetic D) Performing well **Correct Answer: B**
25. Following the chain backwards in the "Healthy Lifestyle" example, what leads to "missing breakfast"? A) Sleeping early **B) Waking up late** C) Feeling tired by noon D) Exercising **Correct Answer: B**
26. Which AI application uses semantic networks to detect emotions in text (sentiment analysis)? A) Expert Systems B) Ontology Development C) Information Retrieval **D) Natural Language Processing (NLP)** **Correct Answer: D**
27. How do semantic networks help expert systems simulate human expertise? A) By providing a fast user interface. **B) By modeling domain knowledge (concepts and relationships) for reasoning.** C) By learning directly from human experts. D) By storing large datasets. **Correct Answer: B**
28. Semantic networks are often used as a foundation for building more formal, structured maps of domain knowledge called: A) Databases B) Algorithms **C) Ontologies** D) Neural Networks **Correct Answer: C**
29. How do semantic networks aid ontology development? A) By providing the reasoning engine. **B) By showing how concepts are related and organizing them hierarchically.** C) By defining the user interface. D) By storing instance data. **Correct Answer: B**
30. In Information Retrieval, semantic networks help search engines understand context. What example is given for searching "healthy meal for the heart"? A) Returning only pages with the exact phrase. **B) Including articles on Mediterranean diet or cholesterol.** C) Showing advertisements for heart medication. D) Ranking results by website popularity only. **Correct Answer: B**
31. Making the decision-making process of Machine Learning models more transparent is aided by semantic networks through enhanced: A) Prediction accuracy B) Training speed **C) Model interpretability** D) Data storage **Correct Answer: C**

32. The ease of understanding and visualizing relationships in semantic networks corresponds to which advantage? A) Flexible Structure B) Contextual Understanding C) Interoperability **D) Intuitive Representation** **Correct Answer: D**
33. The ability to add new concepts and relationships easily relates to which advantage of semantic networks? A) Intuitive Representation **B) Flexible Structure** C) Efficient Inference D) Interoperability **Correct Answer: B**
34. Capturing how word meanings relate to context is part of which advantage? A) Flexible Structure **B) Contextual Understanding** C) Interoperability D) Efficient Inference **Correct Answer: B**
35. Facilitating knowledge sharing between different systems due to a formal structure relates to which advantage? A) Intuitive Representation B) Flexible Structure C) Contextual Understanding **D) Interoperability** **Correct Answer: D**
36. Quicker deduction of new knowledge based on existing relationships relates to which advantage? A) Intuitive Representation B) Flexible Structure **C) Efficient Inference** D) Interoperability **Correct Answer: C**
37. **Performance degradation in large-scale systems** is linked to which disadvantage? A) Complexity in Maintenance B) Limited Expressiveness **C) Scalability Issues** D) Difficulty in Capturing Temporal Relationships **Correct Answer: C**
38. The challenge of keeping the network accurate as concepts evolve relates to which disadvantage? A) Scalability Issues **B) Complexity in Maintenance** C) Limited Expressiveness D) Difficulty in Capturing Temporal Relationships **Correct Answer: B**
39. The inability to easily represent ambiguity or probabilistic relationships falls under which disadvantage? A) Scalability Issues B) Complexity in Maintenance **C) Limited Expressiveness** D) Difficulty in Capturing Temporal Relationships **Correct Answer: C**
40. Struggling to represent time-dependent information is which disadvantage? A) Scalability Issues B) Complexity in Maintenance C) Limited Expressiveness **D) Difficulty in Capturing Temporal Relationships** **Correct Answer: D**
41. Which type of network is also known as a taxonomic network? A) Assertional B) Implicational **C) Definitional** **D) Executable** **Correct Answer: C**
42. Representing the fact "MyCar is located in Cairo" would be done in which type of network? A) Definitional B) **Assertional** C) Implicational D) Executable **Correct Answer: B**

43. Which network type is most concerned with representing cause-and-effect or if-then relationships? A) Definitional B) Assertional C) Implicational D) Executable
Correct Answer: C
44. A network describing the steps to log into a website would be an example of which type? A) Definitional B) Assertional C) Implicational D) Executable
Correct Answer: D
45. A network that automatically adds a new node for "Golden Retriever" under "Dog" after processing text about dog breeds is exhibiting characteristics of which network type? A) Definitional B) Assertional C) Executable D) Learning
Correct Answer: D
46. If a semantic network uses "is-a" links for hierarchy and also contains nodes for specific instances like "Fido", it is likely a: A) Purely Definitional Network B) Purely Assertional Network C) Hybrid Network D) Executable Network
Correct Answer: C
47. What is the primary information conveyed by the edges in a semantic network? A) The size or importance of the concepts. B) The relationships between the concepts. C) The data type of the concepts. D) The author of the concept.
Correct Answer: B
48. The process of determining the specific meaning of a word like "bank" (river bank vs. financial institution) in NLP is called: A) Parsing B) Stemming C) Disambiguation D) Tokenization
Correct Answer: C
49. Which component is NOT explicitly listed as a key component/feature of semantic networks? A) Nodes B) Edges C) Inheritance D) Inference Engine
Correct Answer: D (The inference engine USES the network, but isn't part of its structure)
50. If node A is linked to node B by 2 edges, and node C is linked to node B by 5 edges, which node is considered semantically closer to B based on path length? A) Node A B) Node C C) They are equally close. D) Semantic distance is not related to path length.
Correct Answer: A
51. Ontologies built using semantic networks provide a foundation for which web initiative? A) The World Wide Web Consortium (W3C) B) The Internet of Things (IoT) C) The Semantic Web D) Web 2.0
Correct Answer: C
52. Representing that a "Wheel" is part of a "Car" uses which type of relationship label? A) Is-A B) Has-A C) Instance-Of D) Part-Of
Correct Answer: D
53. Representing that a "Car" has "Wheels" uses which type of relationship label? A) Is-A B) Has-A C) Instance-Of D) Part-Of
Correct Answer: B

54. Which disadvantage highlights that creating good semantic networks requires significant **effort and expertise**? A) Scalability Issues **B) Complexity in Maintenance / Creation** C) Limited Expressiveness D) Difficulty in Capturing Temporal Relationships **Correct Answer: B**
55. If a system needs to represent that an event happened before another event, which disadvantage of semantic networks might become relevant? A) Scalability Issues B) Complexity in Maintenance C) Limited Expressiveness **D) Difficulty in Capturing Temporal Relationships** **Correct Answer: D**
56. The ability to represent that "justice" is a type of "abstract idea" would utilize which core components of a semantic network? **A) Nodes** and an "is-a" labeled edge. B) Only nodes. C) Only edges. D) Attributes and inheritance. **Correct Answer: A**
57. Which type of network would be most appropriate for storing facts like "John owns MyCar" and "MyCar is blue"? A) Definitional **B) Assertional** C) Implicational D) Executable **Correct Answer: B**
58. The inference in the "Healthy Lifestyle" example (e.g., sleeps early -> wakes up early -> ...) relies on chaining what? A) Hierarchical relationships B) Assertions about individuals **C) Implications** D) Procedural steps **Correct Answer: C**
59. Semantic networks are described as being both machine-readable and interpretable by humans. The human interpretability relates most to which advantage? A) Flexible Structure B) Interoperability C) Efficient Inference **D) Intuitive Representation** **Correct Answer: D**
60. Which application area involves using semantic networks to model the knowledge of human experts to aid decision-making? A) Natural Language Processing **B) Expert Systems** C) Ontology Development D) Information Retrieval **Correct Answer: B**

NEW MCQs from: lecture 12_Frames in ES and KR practice question.pdf (pdf4_frames_practice.txt) - Set 2

Instructions: Choose the best answer for each question. These questions are different from the previous set.

1. In frame systems, how does inheritance handle slots not explicitly defined in a child frame? A) The slot remains empty. B) The system prompts the user for a value. **C)** The system checks the parent frame(s) for the value. D) A default procedure is automatically triggered. **Correct Answer: C**
2. Can frame inheritance involve multiple parent frames? A) No, only single inheritance is allowed. **B)** Yes, the text mentions checking parent frame(s), implying single or multiple inheritance is possible. C) Only if the parent frames have no overlapping slots. D) Only in ontology-based frame systems. **Correct Answer: B**
3. Default Value Inference applies when: A) A slot value is inherited. B) A slot value is explicitly defined in an instance. **C)** A slot isn't filled in an instance, and the frame definition provides a default. D) An IF-NEEDED procedure fails. **Correct Answer: C**
4. How does Default Value Inference differ from Inheritance Inference regarding scope? A) Default applies only to parent frames, Inheritance to child frames. **B)** Inheritance applies across frames, Default can apply even within the same frame if a slot isn't filled by an instance. C) Default only works for numerical slots, Inheritance for all types. D) Inheritance is mandatory, Default is optional. **Correct Answer: B**
5. What is the most dynamic and flexible type of inference mentioned for frames? A) Inheritance Inference B) Default Value Inference **C)** Procedural Attachment Inference D) Slot Constraint Checking **Correct Answer: C**
6. What triggers an IF-NEEDED procedural attachment? A) Writing a new value to the slot. B) Deleting the frame instance. **C)** Accessing the slot when its value is missing. D) The frame inheriting from a parent. **Correct Answer: C**
7. What triggers an IF-ADDED procedural attachment? A) Accessing the slot when its value is missing. **B)** Writing a new value into the slot. C) No value being given for the slot. D) The frame being created. **Correct Answer: B**

8. What triggers a DEFAULT procedural attachment (as described alongside IF-NEEDED/IF-ADDED)? A) Accessing the slot when its value is missing. B) Writing a new value into the slot. **C) No value being given for the slot (providing a fallback).** D) The slot value being inherited. **Correct Answer: C**
9. What is a key benefit of procedural attachments like IF-NEEDED? A) They simplify the frame hierarchy. **B) They allow computation (e.g., fetching from a database) to be part of the reasoning process.** C) They enforce static type checking. D) They guarantee faster slot access. **Correct Answer: B**
10. The `validatePrerequisites()` function in the Student/Course IF-ADDED example serves primarily for: A) Calculating the course grade. B) Logging student enrollment. C) Checking if the student meets requirements before enrollment is finalized. D) Assigning a default grade. **Correct Answer: C**
11. If a `Course` frame has a default `FinalExamDuration` of 90 mins, but instance `ML402` explicitly sets it to 120 mins, which value is used for `ML402`? A) 90 mins (Default) B) 120 mins (Explicit override) C) An average of 105 mins D) An error occurs due to conflict **Correct Answer: B**
12. In the TV example, `ElectronicDevice` is a _ frame relative to `TV`. A) Child B) Sibling C) Parent D) Instance **Correct Answer: C**
13. In the TV example, `TVModel_X100` is a(n) _ of `SmartTV`. A) Parent frame B) Slot C) Instance D) Procedural attachment **Correct Answer: C**
14. Which slot is inherited by `TV` from `ElectronicDevice`? A) `screenSize` B) `panelType` C) `powerSupply` D) `hasWiFi` **Correct Answer: C**
15. Which slot is added specifically in the `SmartTV` frame (not inherited from `TV`)? A) `screenSize` B) `panelType` C) `modelName` D) `operatingSystem` **Correct Answer: D**
16. The `calculatePrice()` procedure attached to the `getPrice` slot is an example of: A) Inheritance B) Default value C) IF-NEEDED attachment D) IF-ADDED attachment **Correct Answer: C**
17. The `logComponentAddition()` procedure attached to the `addComponent` slot is an example of: A) Inheritance B) Default value C) IF-NEEDED attachment D) IF-ADDED attachment **Correct Answer: D**
18. When `TVModel_X100` (an instance of `SmartTV`) is accessed for its `warrantyPeriod`, and it's not defined in `TVModel_X100` or `SmartTV` or `TV`, where does the value come from? A) It remains undefined. B) An IF-NEEDED

procedure is triggered. C) The default value (12 months) is inherited from `ElectronicDevice`. D) The user is prompted. **Correct Answer: C**

19. Lazy inference, as demonstrated by the `getPrice` IF-NEEDED example, means: A) The inference process is slow. B) The value is computed only when it is actually needed/accessed. C) The inference engine ignores some rules. D) Default values are preferred over computed values. **Correct Answer: B**
20. What information is logged by `logComponentAddition()` when component `CMP894` from supplier `AudioMax` is added to `Line12`? A) Only the component ID (`CMP894`). B) The component ID, supplier, line, and date/time. C) The price of the component. D) The TV model the component is for. **Correct Answer: B**
21. In the artwork example, `Painting` and `Sculpture` both inherit from: A) Artist B) Exhibition C) Artwork D) Material **Correct Answer: C**
22. Which frame type in the artwork example has slots for `material` and `weight`? A) Artwork B) Painting C) Sculpture D) Artist **Correct Answer: C**
23. Which frame type in the artwork example has slots for `medium` and `style`? A) Artwork B) Painting C) Sculpture D) Exhibition **Correct Answer: B**
24. The `MonaLisa` frame is described as an instance of which frame type? A) Artwork B) Painting C) Artist D) Exhibition **Correct Answer: B**
25. The `David` frame is described as an instance of which frame type? A) Artwork B) Painting C) Sculpture D) Artist **Correct Answer: C**
26. What procedure is attached via IF-NEEDED to the `getEstimatedValue` slot in the `Artwork` frame? A) `logCreation()` B) `logDisplayInfo()` C) `estimateValue()` D) `calculatePrice()` **Correct Answer: C**
27. What procedure is attached via IF-ADDED to the `addArtwork` slot in the `Artist` frame? A) `estimateValue()` B) `logCreation()` C) `logDisplayInfo()` D) `validatePrerequisites()` **Correct Answer: B**
28. What procedure is attached via IF-ADDED to the `addArtwork` slot in the `Exhibition` frame? A) `estimateValue()` B) `logCreation()` C) `logDisplayInfo()` D) `getAvailableRoom()` **Correct Answer: C**
29. The `estimateValue()` procedure likely uses which slots to calculate the value? A) title, artist B) style, age (derived from year), medium C) material, weight D) exhibitID, startDate, endDate **Correct Answer: B**

30. What is the purpose of the `logCreation()` procedure? A) To estimate artwork value. B) To record when an artwork is added to an artist's portfolio. C) To log when an artwork is displayed in an exhibition. D) To validate the artwork's medium.

Correct Answer: B

31. What is the purpose of the `logDisplayInfo()` procedure? A) To record the creation of an artwork. B) To store exhibit history and curatorial notes for displayed artworks. C) To update the artist's nationality. D) To estimate the exhibition's ticket price. **Correct Answer: B**

32. In the artwork query example (

NEW MCQs from: lecture 10 ontologies and frames.pdf (pdf5_ontologies_frames.txt) - Set 2

Instructions: Choose the best answer for each question. These questions are different from the previous set.

1. What is the formal definition of an ontology in knowledge representation? A) A graphical representation of concepts. B) A data structure with slots and fillers. **C) A formal, explicit specification of a shared conceptualization.** D) A collection of rules for expert systems. **Correct Answer: C**
2. Which of the following is NOT typically part of an ontology definition? A) Classes (Concepts) B) Relations (Properties) **C) Procedural Attachments (like IF-NEEDED)** D) Axioms (Rules/Constraints) **Correct Answer: C** (Procedural attachments are more characteristic of Frames)
3. What is the first step in the Ontology Development Process described? A) Define Classes and Instances B) Define Relationships **C) Domain Analysis** D) Add Axioms and Constraints **Correct Answer: C**
4. Creating a hierarchy of classes and defining specific instances falls under which step of ontology development? A) Domain Analysis **B) Define Classes and Instances** C) Define Relationships D) Evaluation and Refinement **Correct Answer: B**
5. Specifying how classes and instances are related (e.g., using properties) occurs in which ontology development step? A) Domain Analysis B) Define Classes and

Instances **C) Define Relationships** D) Add Axioms and Constraints **Correct Answer: C**

6. Defining logical rules to ensure correctness and consistency is part of which ontology development step? A) Define Classes and Instances B) Define Relationships **C) Add Axioms and Constraints** D) Evaluation and Refinement **Correct Answer: C**

7. Continuously checking the ontology for correctness and completeness is which step? A) Domain Analysis B) Define Relationships C) Add Axioms and Constraints **D) Evaluation and Refinement** **Correct Answer: D**

8. How do ontologies primarily promote standardization and interoperability? A) By using a specific programming language. **B) By providing a shared understanding and vocabulary for a domain.** C) By enforcing strict data types. D) By automatically translating between formats. **Correct Answer: B**

9. Ontologies are key components enabling more intelligent information retrieval in which major web initiative? A) Web 2.0 B) The Internet of Things (IoT) **C) The Semantic Web** D) Cloud Computing **Correct Answer: C**

10. How do ontologies support reasoning and inference? A) By storing large amounts of instance data. **B) By defining axioms and relationships that allow deriving new knowledge from existing facts.** C) By providing default values for slots. D) By using graphical nodes and edges. **Correct Answer: B**

11. The example "Simba is a mammal" deduced from "Every lion is a mammal" and "Simba is a lion" illustrates which capability enhanced by ontologies? A) Knowledge Sharing B) Standardization **C) Reasoning and Inference** D) Semantic Search **Correct Answer: C**

12. How do ontologies improve semantic search compared to keyword search? A) By indexing keywords faster. **B) By providing context and understanding meaning, not just matching words.** C) By limiting results to a specific domain. D) By using graphical interfaces. **Correct Answer: B**

13. SNOMED CT and Gene Ontology are examples of ontologies used in which application domain? A) E-commerce B) Robotics C) Natural Language Processing **D) Healthcare** **Correct Answer: D**

14. How are ontologies used in e-commerce? A) To process payments securely. **B) To categorize products for efficient search and recommendation.** C) To design the user interface. D) To manage user accounts. **Correct Answer: B**

15. In robotics, how do ontologies aid robots? A) By providing physical power. **B)** By representing understanding of the environment, objects, and tasks for navigation/decision-making. C) By controlling motor functions directly. D) By translating sensor data into raw signals. **Correct Answer: B**
16. Which advantage highlights that ontologies provide a common vocabulary for systems to communicate? A) Improved Knowledge Sharing B) Enhanced Reasoning **C)** Standardization D) Scalability **Correct Answer: C**
17. Which advantage relates to the ability to extend or refine ontologies as new knowledge emerges? A) Standardization B) Enhanced Reasoning C) Contextual Search **D)** Scalability **Correct Answer: D**
18. What is a major disadvantage related to the effort required to build an ontology? A) Limited Expressiveness B) Performance Overhead **C)** Complexity (resource-intensive, requires deep domain knowledge) D) Update Overhead **Correct Answer: C**
19. Ensuring that adding new concepts doesn't create logical contradictions relates to which disadvantage? A) Complexity **B)** Consistency Challenges C) High Initial Cost D) Limited Expressiveness **Correct Answer: B**
20. The difficulty in representing tacit (implicit or hard-to-formalize) knowledge is which disadvantage? A) Complexity B) Consistency Challenges C) Performance Overhead **D)** Limited Expressiveness **Correct Answer: D**
21. The computational cost associated with reasoning over large ontologies is which disadvantage? **A)** High Initial Cost B) Limited Expressiveness C) Performance Overhead D) Update Overhead **Correct Answer: C**
22. The need for ongoing work to keep the ontology accurate and relevant is which disadvantage? A) Complexity B) Consistency Challenges C) Performance Overhead **D)** Update Overhead **Correct Answer: D**
23. According to the comparison table, which structure uses nodes linked by edges? **A)** Semantic Networks B) Frames C) Ontologies D) All of the above **Correct Answer: A**
(While Ontologies can be visualized as graphs, the primary definition given here for node/edge structure is Semantic Networks)
24. According to the comparison table, which structure is described as a "Structured representation with slots and values"? A) Semantic Networks **B)** Frames C) Ontologies D) Both Frames and Ontologies **Correct Answer: B**

25. Which knowledge representation method is considered the most expressive and supportive of **complex reasoning** and constraints according to the table? A) Semantic Networks B) Frames **C) Ontologies** D) They are equally expressive.
Correct Answer: C
26. Which method is best suited for representing objects with detailed attributes and inheritance, particularly in expert systems? A) Semantic Networks **B) Frames** C) Ontologies D) Rule-based systems **Correct Answer: B**
27. Which method is recommended when you need formalized, structured knowledge with reasoning capabilities for **complex domains** like the Semantic Web? A) Semantic Networks B) Frames **C) Ontologies** D) Databases **Correct Answer: C**
28. What are the core components of a Frame data structure? A) Nodes, Edges, Labels B) Classes, Instances, Relations, Axioms **C) Slots, Fillers, Defaults, Inheritance mechanism** D) Rules, Facts, Working Memory **Correct Answer: C**
29. What do "Slots" in a frame represent? A) Specific values or data. B) Default assumptions. **C) The attributes or properties of the object/concept.** D) Links to parent frames. **Correct Answer: C**
30. What do "Fillers" in a frame represent? A) The attributes or properties. B) Default assumptions. **C) The specific values or data that populate the slots.** D) Inheritance links. **Correct Answer: C**
31. What is the role of "Defaults" in a frame? A) To define the frame's name. **B) To provide values for slots when no specific value is given (can be overridden).** C) To link to other frames. D) To specify the data type of a slot. **Correct Answer: B**
32. What is the first step described for how frame inheritance works? A) Create Child Frame **B) Define Parent Frame** C) Use Inherited Attributes D) Override or Extend **Correct Answer: B**
33. When a child frame is created, what does it automatically get from its parent? A) Only the parent's name. **B) All slots and values (unless overridden).** C) Only the default values. D) A unique identifier. **Correct Answer: B**
34. Can a child frame change or add to the attributes it gets from a parent frame? A) No, inherited attributes are fixed. **B) Yes, it can override inherited values and add its own unique slots (Override or Extend).** C) It can only add new slots, not change inherited ones. D) It can only change inherited slots, not add new ones. **Correct Answer: B**

35. In the Car example, "Vehicle" is the _____ frame. A) Child B) Instance **C) Parent** D) Sibling **Correct Answer: C**
36. In the Car example, "Car" inherits slots like "Make" and "Model" from which frame? A) It defines them itself. B) From the "Engine" frame. **C) From the "Vehicle" frame.** D) From the "Transmission" frame. **Correct Answer: C**
37. In the Car example, "Number of Doors" is a slot that is: A) Inherited from "Vehicle". **B) Added specifically in the "Car" frame.** C) A default value. D) Overridden from "Vehicle". **Correct Answer: B**
38. In the final Car example summary, the "Year" slot has the value 2023. This value is described as: A) Inherited from Vehicle. B) Added in Car. C) A default value. **D) Overridden (from the implicit parent value, or just specified in the child).** **Correct Answer: D**
39. In the Animal Classification example, which frame is the most general? **A) Animal** B) Mammal C) Lion D) They are all at the same level. **Correct Answer: A**
40. In the Animal Classification example, which frame is the most specific? A) Animal B) Mammal **C) Lion** D) They are all at the same level. **Correct Answer: C**
41. The "Mammal" frame inherits attributes like "Habitat" and "Diet" from which frame? A) Lion **B) Animal** C) It defines them itself. D) Species **Correct Answer: B**
42. The attribute "Warm-Blooded" is added in which frame in the Animal example? A) Animal **B) Mammal** C) Lion D) It's inherited from a higher level. **Correct Answer: B**
43. The attribute "Mane Size" is specific to which frame in the Animal example? A) Animal B) Mammal **C) Lion** D) Habitat **Correct Answer: C**
44. What is the inheritance relationship shown for the Lion frame? A) Lion → Animal → Mammal B) Animal → Mammal → Lion **C) Lion → Mammal → Animal** D) Mammal → Lion → Animal **Correct Answer: C**
45. In the Fitness Class example, "fitness class" is described as a specialization of: A) instructor **B) group activity** C) workout D) gym studio **Correct Answer: B**
46. In the Fitness Class example, "yoga mat" has a superclass of: A) fitness class **B) equipment** C) foam D) floor-based exercises **Correct Answer: B**
47. In the Fitness Class example, "cardio workout" and "strength workout" are specializations of: A) fitness class B) instructor **C) workout** D) dumbbells **Correct Answer: C**

48. What information is stored in the "contains" slot of the "fitness class" frame? A) The location and duration. B) The specialization type. **C) The components like instructor, equipment, and workout.** D) The goal of the class. **Correct Answer: C**
49. The frame description for "Hotel Bed" shows it inherits from "Bed" and adds slots like: **A) size, firmness** B) material, color C) room_number, floor D) length, width **Correct Answer: A**
50. The frame description for "Hotel Room" includes slots for components like: A) lobby, restaurant, pool B) bed, chair, bath, phone C) check_in_time, check_out_time D) price, availability **Correct Answer: B**
51. In NLP, how might frames be used to understand a reservation request? A) By translating the request into different languages. B) By identifying the sentiment of the request. **C) By using a "booking" frame to extract key details** (date, time, number of people). D) By correcting grammatical errors. **Correct Answer: C**
52. How do expert systems utilize frames? A) To create the user interface. **B) To structure domain-specific knowledge** (e.g., diseases, symptoms, treatments). C) To perform complex mathematical calculations. D) To store user login credentials. **Correct Answer: B**
53. In robotics, frames help robots understand their environment by: A) Providing power to sensors. **B) Representing objects and their characteristics in a structured way.** C) Executing movement commands directly. D) Communicating with other robots. **Correct Answer: B**
54. Which knowledge representation focuses on defining concepts, entities, and relationships within a domain using classes, instances, properties, and axioms? A) Semantic Networks B) Frames **C) Ontologies** D) Rule-Based Systems **Correct Answer: C**
55. Which knowledge representation uses slots, fillers, defaults, and inheritance to represent stereotyped situations or objects? A) Semantic Networks **B) Frames** C) Ontologies D) Production Rules **Correct Answer: B**
56. If you need to represent that a specific car instance, "MyCar", has the color "Blue", which frame component would hold "Blue"? A) Slot (e.g., Color) **B) Filler** (for the Color slot) C) Default D) Inheritance link **Correct Answer: B**
57. The ability of a "Lion" frame to automatically possess the "Warm-Blooded: Yes" property from the "Mammal" frame is due to: A) Default values B) Axioms **C) Inheritance** D) Slots **Correct Answer: C**

58. Which representation is generally considered less formal but more intuitive for simple relationships? **A)** Semantic Networks B) Frames C) Ontologies D) First-Order Logic **Correct Answer: A**
59. Which representation provides a strong foundation for the Semantic Web due to its formal structure and reasoning capabilities? A) Semantic Networks B) Frames **C)** Ontologies D) Mind Maps **Correct Answer: C**
60. If you are building an expert system to diagnose car problems, representing different car models with their specific parts and common issues would likely be best done using: A) Semantic Networks (for basic part connections) B) Frames (to represent each model with its specific slots/attributes and inherit from general car types) C) Ontologies (if very formal reasoning about car types and diagnostic rules is needed) **D)** Primarily Frames, potentially supported by rules or an ontology. **Correct Answer: D** (Frames are well-suited for object representation, often combined with other methods)